TITLE: **Accuracy of naive summation: error growth, conditioning and better algorithms**

SUBJECT: Aim, method and results of the analyses in version 2 of the `Naive-Summation` repository

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1. Aim and scope

Naive (recursive) summation computes $S = \sum_{i=1}^{n} x_i$ as $s_0 = 0$, $s_k = s_{k-1} + x_k$. In floating-point arithmetic every addition is rounded, $\mathrm{fl}(a + b) = (a + b)(1 + \delta)$ with $|\delta| \le \varepsilon$, where $\varepsilon = 2^{-53} \approx 1.1 \times 10^{-16}$ is the unit roundoff of double precision. The computed sum $\hat{S}$ therefore has an error $E = \hat{S} - S$. Version 1 of this project measured E for random data and concluded that naive summation is "numerically unstable and unreliable". This note asks three narrower questions, each answered by one analysis:

1. **How does the error grow with n,** compared with the theoretical bound, and does that depend on the data?
2. **What makes the relative error $|E|/|S|$ large?** Version 1 found that it "fluctuates".
3. **What do better algorithms gain, and at what cost?** And how accurate was the reference sum of version 1?

2. Method

- **Exact errors.** Every error is computed exactly: $\hat{S}$ and $-x_1, \ldots, -x_n$ are added with `math.fsum`, which tracks the sum without intermediate rounding, and the result is rounded once. Comparing with a rounded reference sum instead would add up to half a unit in the last place to every error, which is as large as the error of the best algorithms. The unit tests check this against exact rational arithmetic.
- **Algorithms.** Naive summation (computed with `np.cumsum`, which the tests show is bit-for-bit the same as a Python loop and gives all partial sums at once); pairwise summation, which adds neighbours level by level; and Kahan's (1965) compensated summation, which carries the low-order part lost in each addition into the next.
- **Bounds** (Higham, 2002, ch. 4), with $\gamma_k = k\varepsilon/(1 - k\varepsilon)$: naive $|E| \le \gamma_{n-1} \sum |x_i|$; pairwise $|E| \le \gamma_{\lceil \log_2 n \rceil} \sum |x_i|$; Kahan $|E| \le (2\varepsilon + O(n\varepsilon^2)) \sum |x_i|$. Every error measured in the unit tests stays within these bounds.
- **Data.** Uniform random numbers as in version 1, $\mathrm{U}(-1, 1)$, and for

comparison $\mathrm{U}(0,1)$, where all terms have the same sign. Seeds are fixed.

3. Analysis 1: Error growth (Figure 1)

Why. The bound for naive summation grows linearly in n, and version 1 described the error as growing linearly. But a bound is a worst case: it assumes that every rounding error has the largest size and the same sign. The question is how the error grows in practice.

Method. For each distribution, 30 arrays of 10^6 values; for 26 lengths n between 10 and 10^6 the error on the first n values is computed exactly. The figure shows the root mean square (RMS) of $|E|/(\varepsilon \sum |x_i|)$; in these units the naive bound is the line $n-1$.

How to read the figure. Both axes are logarithmic, so a quantity that grows like n^k appears as a line of slope k. The distance between a curve and the dashed line shows how pessimistic the bound is.

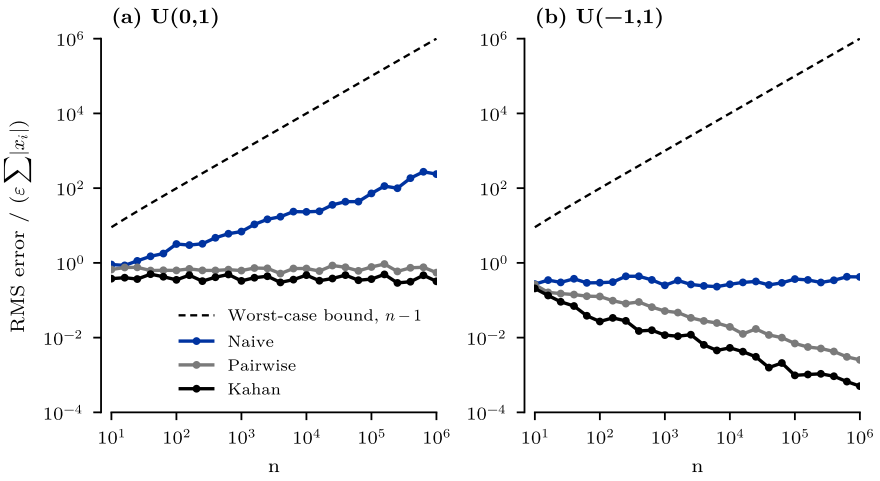


Figure 1. RMS error of three algorithms over 30 arrays, in units of $\varepsilon \sum |x_i|$. Dashed: worst-case bound for naive summation.

Finding. With positive terms, $\mathrm{U}(0,1)$, the naive error grows with slope 0.50 on this scale: like $\sqrt{n}$, as predicted when rounding errors behave like independent random signs (Johnson, 2019). At $n = 10^6$ it is 237 units, against 10^6 for the bound. With terms of both signs, $\mathrm{U}(-1,1)$, the slope is only 0.06: the partial sums stay small, so each rounding error is small, and the error stays below one unit (0.42 at $n = 10^6$). In both cases pairwise and Kahan summation stay below one unit at every n (at $n = 10^6$: 0.55 and 0.32 for $\mathrm{U}(0,1)$; 0.0025 and 0.0005 for $\mathrm{U}(-1,1)$).

Interpretation. "The error grows linearly" describes the bound, not the typical behaviour. How fast the error actually grows depends on the data, through the size of the partial sums.

4. Analysis 2: Conditioning (Figure 2)

Why. The relative error of version 1 varied strongly from run to run. Dividing the naive bound by $|S|$ gives $|E|/|S| \leq \gamma_{n-1}\,\kappa$ with $\kappa = \sum|x_i|/|\sum x_i|$, the condition number of the sum. When terms of both signs nearly cancel, $|S|$ is small and κ is large, so even a small absolute error is a large relative error. This analysis checks whether κ explains the variation.

Method. 2,000 arrays of $n = 10^4$ values from U$(-1, 1)$; for each, κ and the exact relative error of naive and Kahan summation.

How to read the figure. Each point is one array. Points that rise to the right mean that the relative error follows the condition number.

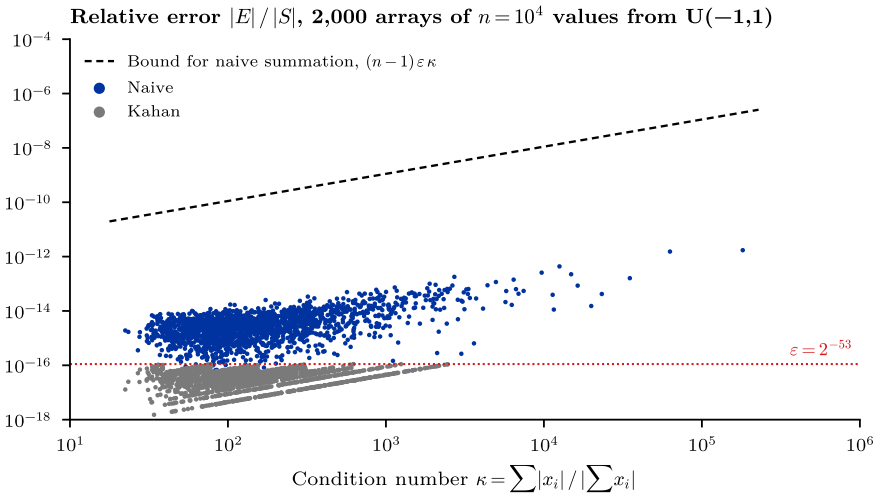


Figure 2. Relative error against the condition number for 2,000 arrays of 10^4 values from U$(-1, 1)$. Dashed: bound for naive summation. Dotted: ε.

Finding. The median κ is 125, the largest 181,407. The naive relative error rises with κ (slope 0.64 on the log–log scale) and stays far below the bound (0 violations). Its mean, 7.6×10^{-15}, is 2.8 times its median, 2.7×10^{-15}, because the 1% of arrays with the largest errors account for 37% of the mean: an average of relative errors, as in version 1, is dominated by a few sums close to zero. Kahan's relative error never exceeds $1.00\,\varepsilon$, whatever κ; its points form bands because the result is within one rounding of the exact sum, and roundings come in discrete steps.

Interpretation. The "fluctuation" is not instability of the algorithm but the conditioning of the problem. Naive summation is backward stable: $\hat{S}$ is the exact sum of inputs perturbed by at most γ_{n-1} in relative terms (Higham, 2002, ch. 4; Johnson, 2019). A large relative error in an ill-conditioned sum is what any backward stable method can

produce.

5. Analysis 3: Algorithms and the version 1 reference (Table 1)

Why. A more accurate algorithm is only worth using if its cost is acceptable, so accuracy and time are measured together. Version 1 used `np.sum(x, dtype=np.longdouble)` as the exact sum; the precision of `longdouble` varies across platforms (NumPy documentation), so its accuracy is measured as well.

Method. 10 arrays of 10^6 values from $\mathrm{U}(-1, 1)$, as in version 1. Errors as in analysis 1; time is the median of five calls (Darwin arm64, Python 3.13.6, NumPy 2.4.1).

Table 1. RMS error in units of $\varepsilon \sum |x_i|$ and running time for $n = 10^6$.

Method	RMS error	Time (ms)
Naive, Python loop (version 1)	0.442	17.86
Naive, `np.cumsum`	0.442	2.09
Pairwise	1.9×10^{-3}	0.47
`np.sum` (NumPy, partial pairwise)	2.4×10^{-3}	0.13
Kahan, Python loop	4.2×10^{-4}	67.10
`math.fsum` (correctly rounded)	4.2×10^{-4}	15.13
`np.sum`, `longdouble` (version 1 reference)	2.4×10^{-3}	0.13

Finding. The Python loop of version 1 is both inaccurate and slow: it takes 17.9 ms, while `math.fsum`, whose result is the exact sum correctly rounded, takes 15.1 ms (1.2 times faster) and `np.sum` takes 0.13 ms with an error two orders of magnitude smaller. Kahan's method reaches the accuracy of `fsum` but is slow when written in Python (67 ms); its advantage is that it needs only a constant amount of extra work per term, which matters in compiled code. On this machine `longdouble` has the same precision as `float64`, so the version 1 reference is simply NumPy's pairwise sum in double precision; its error is 0.5% of the error of naive summation. Measuring errors against an exact reference, as done here, avoids this platform dependence.

6. Conclusions and limitations

- The error of naive summation is far below its worst-case bound. It grows like $\sqrt{n}$ on the bound's scale for terms of one sign and hardly grows for terms that cancel. The relative error is governed by the condition number.
- Naive summation is backward stable, not unstable. For accurate sums in Python, `math.fsum` is both correctly rounded and faster than a Python loop; `np.sum` is fast and much more accurate than naive summation.
- Only uniform data, double precision and sizes up to 10^6 were studied.

Times were measured on a single machine.

References

Higham, N. J. (1993). The accuracy of floating point summation. *SIAM Journal on Scientific Computing*, 14(4), 783–799.

Higham, N. J. (2002). *Accuracy and Stability of Numerical Algorithms* (2nd ed.). SIAM.

Johnson, S. G. (2019). Notes on the accuracy of naive summation; and Backwards stability of recursive summation. MIT course 18.335J, *Introduction to Numerical Methods*, Lecture 3.

Kahan, W. (1965). Further remarks on reducing truncation errors. *Communications of the ACM*, 8(1), 40.

NumPy documentation: `numpy.sum`, `numpy.cumsum`, `numpy.finfo`. Python documentation: `math.fsum`.

All numbers are generated from `figures/*.csv` by `make_paper.py`.